

Sanatan Khemariya

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PROFESSIONAL SUMMARY

AI/Backend Engineer shipping production AI systems — LLM-powered support engines, financial research platforms, and voice agents. Strong focus on RAG pipelines, cost optimization, and backend reliability.

EDUCATION

Jaypee University of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

Guna, India

Aug. 2022 – Jul. 2026

IIT Guwahati

Credit-Linked Program in Data Science

Guwahati, India

Jan. 2025 – Jan. 2026

PROFESSIONAL EXPERIENCE

AI/Backend Engineer

PetRay Pvt. Ltd

Jan 2026 – Present

Remote

- Shipped a **FastAPI + PostgreSQL** AI support microservice for Urban Gabru's D2C brands (multi-tenant Shopify) — AI Draft-Reply with risk flags, LLM tool-calling for policy fetch, and a batch auto-reply feature with hard escalation guardrails; prompt-caching cut input cost by over half
- Deployed on Docker + GitHub Actions CI/CD with dual-env DB (Supabase dev / AWS RDS prod), structured logging, Alembic migrations, and a full pytest suite
- Building a **real-time voice agent** for Hindi/Hinglish/English calls using Gemini Live API + Cartesia TTS with barge-in handling and Vobiz telephony (in progress)

AI Engineer → Founding Engineer

BharatNiti AI

May 2025 – Jan 2026

Remote

- Architected a **financial research analyst** for 6,262+ Indian companies — FastAPI backend with **10× improvement** over traditional RAG via semantic query expansion and multi-query processing
- Built multi-layer caching (PostgreSQL + SQLite fallback) achieving **95%+ cache hit rate**; parallel doc processing cut ingestion from 10–20 min to **2–5 min**; real-time cost tracking across OpenAI/Vertex AI

PROJECTS

Azazel – Legacy Model Enhancement | *LLMs, OpenAI, FAISS, LangChain, FastAPI* | [Live](#) | [GitHub](#)

- Transformed legacy LLMs into multimodal assistants (text, image, code, web search) with a FastAPI streaming backend at sub-100ms latency; reduced API costs by **30%** via prompt optimization and response caching

Legal Document Analyzer | *BART, Transformers, PPO RL, Hugging Face* | [Link](#) | [GitHub](#)

- Fine-tuned **facebook/bart-large-cnn** on legal datasets; built custom transformer with BPE (50k vocab) demonstrating multi-head attention for domain-specific summarization

Easy-Notes – AI Writing Assistant | *FastAPI, TensorFlow, Transformer, Supabase* | [GitHub](#)

- Full-stack note-taking platform with LSTM next-word prediction (**66% accuracy**) and Transformer English-to-Hindi translation (**94.6% accuracy**); served under 100ms latency via FastAPI

TECHNICAL SKILLS

Languages: Python, C++, Swift

Frameworks: FastAPI, LangChain, OpenAI SDK, PyTorch, TensorFlow, Scikit-learn

Databases: PostgreSQL, MySQL, Vector Databases (FAISS)

Cloud & DevOps: AWS (RDS, S3, EC2, SSM), Supabase, Docker, GitHub Actions, Google Vertex AI

AI/ML Tools: Hugging Face, Gemini Live API, Cartesia, Google Colab

ACHIEVEMENTS & CERTIFICATIONS

TATA Innovant GenAI Hackathon Finalist — Led the only selected team from Madhya Pradesh among 2,600+ teams

Kaggle — Pandas for Data Analysis | [Link](#) **Kaggle** — Generative AI Intensive | [Link](#)

LeetCode: 300+ problems solved | [Profile](#)